A Hybrid Machine Learning Framework for Detecting and Forecasting Volatility Spillovers Between U.S. and Indian Equity Markets.

Contents

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Executive Summary

This project develops a volatility spillover detection and forecasting framework integrating econometric models (HAR and GARCH). Machine learning is further used to predict next one day values. Using 15 years of daily data across NIFTY, SPX, and VIX (2010–2025), the study replicates well-documented academic results on volatility persistence, clustering, and cross-market transmission, and extends them by building a hybrid predictive model suitable for real-world emerging market conditions.

The analysis reveals strong and regime-dependent spillover effects, with SPX weekly realized volatility and VIX shocks emerging as significant predictors of next-day NIFTY volatility. Rolling correlations also show heightened co-movement during global stress period of covid, confirming that Indian market volatility is influenced not only by domestic factors but also by U.S. risk dynamics.

Forecasting performance demonstrates that a hybrid HAR+GARCH+ML architecture outperforms individual models, reducing out-of-sample error and capturing both long-memory and short-run volatility components. This validates the practical value of combining structural econometric features with nonlinear machine learning methods. Overall, the project contributes a robust, interpretable, and deployable volatility engine that detects spillovers.

Motivation Behind the Project

As an individual trader in the market, I find that NIFTY index is increasingly integrated into global capital flows, which means volatility here is no longer purely domestic. Major U.S. markets specifically the S&P 500 and the VIX have become important external drivers. Understanding this transmission is essential for:

- Building trading strategies
- Deploying hedges to manage open exposures

What is Volatility, understanding its nature and reasons why one should forecast volatility

In finance volatility can be defined as a measure of how quickly the price changes in a given period of time. High volatility means prices fluctuate dramatically, while low volatility indicates more stable, steadier prices.

Nature of Volatility

Volatility exhibits 3 characteristics – Clustering, Persistence & Spillovers:

- Clustering – Volatility Clustering means large price movements tend to follow large movements regardless of direction. In our model alpha + beta ~ 0.98 showing yesterday's shock in SPX or VIX strongly influenced Nifty
- Persistence – Volatility is not short lived i.e. volatility doesn't just depend on yesterday it depends on patterns accumulated over days, weeks, months. This is also called as called long memory
- Spillover effect - Volatility spillover refers to the transmission of risk or shocks from one market into another. When the U.S. becomes volatile, global markets especially emerging ones like India also become more volatile.

Why should one forecast volatility

Forecasting volatility is crucial because it is a *primary input* into risk models, derivatives pricing models (like Black-Scholes), margin requirements, optimal portfolio weights (e.g., volatility-targeting), position sizing rules. Misestimating volatility results in under-hedging, mispriced options, and portfolio drawdowns.

Data Collection and Processing

Data Collection was done by an api call from Yahoo Finance api " yfinance". The data directly pulled was from 2010 to 2025 adjusted close prices of Nifty, SPX and VIX. Macro-event calendars (FOMC meetings, CPI releases) were manually compiled and merged as event-window indicators (e.g., ±2-day windows).

Data Split

70% of data was split into model training i.e. 2010 -18.

15% into validation 2019 -21.

Balance 15% into test data 2021- 25.

Data Preprocessing

- Outlier Handling – The outlier returns exhibited during global crisis of 2020 were retained post inspection as they provided valuable insights into the volatility spiks during those periods
- Transformations - Log (RV) was used to stabilize variance and reduce right skewness. Weekly RVs used rolling-window averages.

Why Nifty and SPX

NIFTY and S&P 500 were chosen because they represent two economically important but structurally different markets: NIFTY as a major emerging market index, and SPX as the world's most influential benchmark. India is increasingly integrated with global capital flows, and U.S. market volatility often transmits into emerging markets through foreign investment, macro shocks, and global risk sentiment.

By studying NIFTY–SPX dynamics, we can measure how much of India's volatility is domestically driven versus globally imported

Why HAR- RV and GARCH were chosen

We used HAR-RV and GARCH because they capture two different and complementary characteristics of volatility. HAR-RV models long-memory behavior by using daily, weekly, and monthly realized volatility, which reflects how volatility decays slowly over multiple horizons. GARCH, on the other hand, captures short-term volatility clustering — the tendency for shocks today to strongly influence volatility tomorrow.

Since real-world volatility contains both long-term persistence and short-term clustering, combining HAR and GARCH allows us to model the full dynamic structure of volatility.

Empirically, combining these two explains more variation than either alone — a finding consistent with research in Andersen–Bollerslev (2003), Corsi (2009), and Audrino–Bühlmann (2015).

Why Machine learning using Random Forests

Random Forests because they strike the right balance between predictive performance, robustness, and interpretability which is essential when the goal is not only to forecast volatility but also to understand the structure of spillovers across markets.

Volatility data is extremely noisy, exhibits regime shifts, and contains nonlinear interactions. Random Forests handle this well because they average many independent decision trees, which significantly reduces variance and prevents overfitting especially during high-volatility stress episodes. In contrast, boosting methods like XGBoost sequentially fit trees and tend to chase noise unless carefully tuned, making them more fragile when volatility spikes or structural breaks occur.

Another major reason is interpretability. This project aims to detect whether SPX and VIX volatility meaningfully contribute to NIFTY volatility. Random Forests provide stable and intuitive feature importance measures, allowing us to directly quantify the predictive contribution of spillover variables. XGBoost, while powerful, offers less transparent importance metrics and often biases toward variables with higher cardinality or proxy effects, making economic interpretation harder.

Finally, Random Forests require fewer hyperparameters and minimal tuning, which is ideal in research environments where robustness matters more than squeezing out an extra 1–2% accuracy. We wanted a model that behaves reliably across different market regimes rather than one that requires intensive calibration and may overfit specific time periods.

So the choice was deliberate: Random Forests give us a stable, interpretable, and resilient ML engine that complements our econometric models and supports the broader research goal of understanding volatility transmission not just predicting numbers.

About the Models

1. HAR-RV Model (Heterogeneous Autoregressive Realized Volatility)

The first component of our framework is the HAR-RV model. Volatility is known to exhibit long-memory not just dependence on yesterday, but on patterns accumulated over days, weeks, and even months.

The HAR model captures this by using realized volatility at multiple horizons:

- daily RV
- weekly RV
- monthly RV

This structure allows us to replicate the slowly decaying persistence documented in empirical finance.

In our results, weekly NIFTY RV emerged as one of the strongest predictors of next-day volatility, confirming the presence of long-memory. Additionally, **weekly SPX RV** was also significant, giving us early evidence of cross-market spillovers.

2. GARCH(1,1) Model (Short-Run Volatility Clustering)

The second component is the GARCH model, which complements HAR by capturing a different property of volatility: **clustering**.

Markets tend to experience turbulence in burst high-volatility days cluster together. GARCH models this short-term dependence through the combination of:

- yesterday's squared return (the shock), and
- yesterday's conditional volatility (the persistence).

GARCH(1,1) estimates showed $\boldsymbol{\alpha + \beta \approx 0.98}$, indicating extremely persistent volatility — very much in line with classic findings in Engle and Bollerslev's literature. This model provides us with a fast-reacting component of volatility: the conditional variance σ_t, which becomes a valuable feature in the ML stage.

3. Spillover Modeling (SPX and VIX → NIFTY)

The next layer of the architecture focuses on detecting cross-market spillovers. India is deeply connected to U.S. financial markets through foreign portfolio flows and global risk sentiment.

We quantified spillovers using:

- SPX daily RV
- SPX weekly RV
- ΔVIX (daily changes in implied volatility)
- FOMC and CPI event windows

Three different methods—HAR coefficients, ML feature importance, and rolling correlations—consistently showed that **U.S. volatility transmits into India's equity market**. For example, ΔVIX had significant predictive power, and SPX weekly RV ranked high in ML importance, demonstrating that NIFTY does not evolve independently but is influenced by global volatility regimes.

4. Hybrid ML Random Forest using HAR + GARCH

Finally, the hybrid model integrates all these components together.

Each model plays a specific role:

- **HAR** captures long-term structural persistence
- **GARCH** captures immediate reactions to shocks
- **ML** captures nonlinear interactions and combines all signals

Because real-world volatility is a mixture of slow and fast components, plus global influences, no single model can fully describe it. The hybrid model produced the **best out-of-sample performance**, validating our hypothesis that volatility is inherently multi-layered and globally interconnected.

Model Performance

Model	Train RMSE	Train MAE	Validation RMSE	Validation MAE	Test RMSE	Test MAE
HAR–ML	2.1749	1.6341	2.3186	1.8072	2.5135	1.7877
GARCH–ML	2.2163	1.6650	2.3381	1.8321	2.5117	1.8453
Hybrid (HAR + GARCH) ML	**2.1469**	**1.6104**	**2.2941**	**1.7841**	**2.4951**	1.8121

The hybrid model reaches a test RMSE of **2.4951**, outperforming:

- HAR–ML (2.5135)
- GARCH–ML (2.5117)

Though the numerical margins appear small, the improvement is *structural*, meaning the information set is richer and the model generalizes better. In volatility forecasting, a reduction of even **0.01–0.02 RMSE** can translate into:

- better hedging ratios
- reduced VaR breaches
- more stable option Greeks
- improved trading strategy risk-adjusted returns

This validates that - **Volatility forecasting is multi-dimensional and requires hybrid models that blend econometrics with machine learning.**

Spillover Regression

RV^{NIFTY}_{t+1} = alpha + beta_1 RV^{NIFTY}_{t} + beta_2 RV^{NIFTY}_{t}^(week) + beta_3 RV^{SPX}_{t} + beta_4 RV^{SPX}_{t}^(week) + beta_5 Delta VIX_t + beta_6(Event Windows) + E

To formally test whether U.S. volatility spills into Indian markets, we estimated a spillover regression using HAR-style domestic terms and SPX/VIX external drivers. The results were economically meaningful and statistically consistent with global risk transmission theory.

Weekly SPX realized volatility emerged as a significant predictor of next-day NIFTY volatility, indicating that medium-term U.S. volatility conditions propagate into India with measurable strength. Additionally, changes in the VIX were significant, highlighting that shifts in global risk sentiment have immediate effects on NIFTY's volatility.

The combination of SPX RV, VIX shocks, and domestic long-memory components explained a substantial portion of NIFTY’s volatility dynamics, confirming that India is strongly linked to global volatility cycles. These results align with the broader spillover literature and motivate the choice to integrate these features into the final machine learning model

Further scope of improvements

While the current hybrid model combining HAR, GARCH, spillovers, and Random Forests performs well, there are several improvements and extensions that could further strengthen the forecasting framework:

- Incorporating multi-time frequency data like intraday data
- Introducing high frequency data
- Extending GARCH to asymmetric models
- Adding option implied information like IV and change in IV
- Deploying advanced ML techniques like XGBoost
- Extending the model to real time analysis mode

Conclusion

This project shows that NIFTY volatility is driven by a combination of domestic long-memory effects, short-run shock clustering, and meaningful spillovers from U.S. markets. HAR captured the persistent structure of volatility, GARCH modeled immediate shock reactions, and the spillover regression confirmed that SPX weekly volatility and VIX shocks significantly influence India’s volatility dynamics.

When these structural insights were combined in a hybrid Random Forest model, we achieved the strongest out-of-sample performance, demonstrating that volatility is best forecasted using a multi-model approach rather than any single framework. The results replicate established findings in the volatility literature and translate them into a practical, data-driven forecasting engine suitable for real-world risk management.

Overall, the project highlights that integrating econometric models with machine learning provides a richer and more accurate understanding of volatility in globally connected markets, and it sets the stage for future extensions using asymmetric models, regime-switching, and higher-frequency data.